React Assessment Paper

Duration: 2 Hours
Total Marks: 50

Section A – MCQs (20 Marks)
(10 Questions × 2 Marks each)

1. What is the purpose of Axios in a React application?
   A. Styling components
   B. Routing pages
   C. Making HTTP requests
   D. Managing state

2. Which hook is used to perform side effects like API calls?
   A. useState
   B. useEffect
   C. useContext
   D. useReducer

3. What does the dependency array in useEffect control?
   A. Component rendering
   B. When the effect runs
   C. API response
   D. State update

4. Which method is used to fetch data using Axios?
   A. axios.push()
   B. axios.fetch()
   C. axios.get()
   D. axios.retrieve()

5. What is the correct way to update state in React?
   A. Direct assignment
   B. Using setState/useState setter
   C. Using global variables
   D. Using DOM manipulation

6. Which HTTP method is typically used to update a record?
   A. GET
   B. POST
   C. PUT
   D. DELETE

7. What is the purpose of key in list rendering?
   A. Styling
   B. Identifying elements uniquely
   C. API handling
   D. Event binding

8. Which hook is used for navigation in React Router?
   A. useRoute
   B. useNavigate
   C. useHistory
   D. useLocation

9. What does Axios return?
   A. JSX
   B. Promise
   C. String
   D. Object only

10. Which lifecycle is replaced by useEffect?
    A. constructor
    B. render
    C. componentDidMount
    D. shouldComponentUpdate

Section B – Short Answer (10 Marks)
(5 Questions × 2 Marks each)

1. What is CRUD in React applications?
2. Explain the difference between controlled and uncontrolled components.
3. What is Axios interceptor?
4. How do you handle errors in Axios?
5. What is the purpose of useParams in routing?

Section C – Coding (20 Marks)

Question 1: Fetch and Display Data (10 Marks)

Create a React component to:

- Fetch users from API: https://jsonplaceholder.typicode.com/users
- Display Name and Email
- Use Axios
- Use useEffect

Question 2: CRUD Operation – Create & Delete (10 Marks)

Build a simple Employee Management Component:

Requirements:

- Input fields: Name, Role
- Add employee (POST request using Axios)
- Display employee list
- Delete employee (DELETE request)
- Use functional components with hooks

Bonus:

- Show success/error messages
- Basic validation